Problem 5

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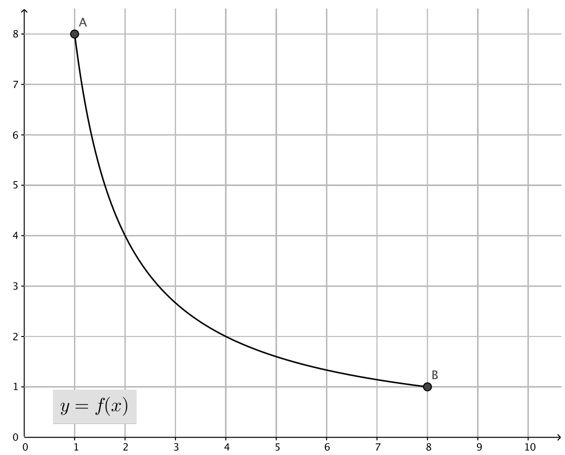
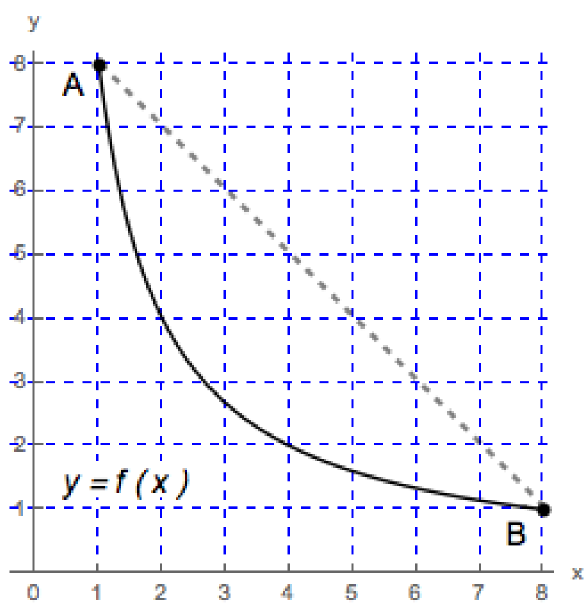
Problem 1 *A curve is given in the figure below, where $f(x) = 8/x$.*



(a) *In the figure above, draw a secant line joining the points $A = (1, f(1))$ and $B = (8, f(8))$.*

Explanation.



(b) *Find the slope, m_{sec}, of this secant line.*

Explanation.

$$m_{\text{sec}} = \frac{f(8) - f(1)}{8 - 1} = \frac{1 - 8}{7} = \frac{-7}{7} = -1$$

(c) *Show that the function f satisfies the conditions of the Mean Value Theorem on the interval $[1, 8]$ and find a point (or points) guaranteed to exist by the Mean Value Theorem.*

Explanation. f is continuous on the interval $[1, 8]$, f is differentiable on the interval $(1, 8)$. By the Mean Value Theorem there exists a(t least one) point c in $(1, 8)$ such that

$$f'(c) = \frac{f(8) - f(1)}{8 - 1} = -1$$

Now $f'(x) = -8/x^2$ hence

$$\begin{aligned} f'(c) = -1 &\iff \frac{-8}{c^2} = -1 \\ &\iff c^2 = 8 \\ &\iff c = \pm 2\sqrt{2} \end{aligned}$$

Since c must be in $(1, 8)$ we have that $c = 2\sqrt{2}$ is the only point in $(1, 8)$ with $f'(c) = -1$.